

- ① Given a curve $y = 7x - x^3$, and x increases at the rate 2 units per second. The rate at which the slope of the curve is changing, when $x = 5$ is: A) -60 units B) -70 units C) 60 units D) -140 units
- ② The order of the differential equation $\frac{d^4 y}{dx^4} - \sin\left(\frac{d^2 y}{dx^2}\right) = 5$ is
A) 4 B) 3 C) 2 D) Not defined
- ③ $\int \frac{1}{x(\log x)^2} dx$ is
- ④ A function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 - 4x + 5$ is
(i) injective but not surjective
(ii) surjective but not injective
(iii) both injective and surjective
(iv) neither injective nor surjective.
- ⑤ If the inverse of matrix $\begin{bmatrix} 7 & -3 & -3 \\ -1 & \lambda & 0 \\ -1 & 0 & 1 \end{bmatrix}$ is $\begin{bmatrix} 1 & 3 & 3 \\ 1 & \lambda & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then find λ .
- ⑥ Derivative of $e^{\sin^2 x}$ with respect to $\cos x$ is.
- ⑦ Find the matrix A^2 , where $A = [a_{ij}]$ is a 2×2 matrix whose elements are given by $a_{ij} = \max(i, j) - \min(i, j)$.
- ⑧ The function $f(x) = \frac{x}{2} + \frac{2}{x}$ has a local minima at $x = ?$